GinCol Lab · Aula

Student worksheet

Musical Language and Practice · 1st year of Bachillerato · A song, layer by layer · Class 1 of 8 · With Malik

What is a song made of?

Class video: https://youtu.be/fcgd5BGAEiw

The code leads to the video.

Name: _____ Date: _____

Before you start

1. You need paper and a pencil. If you can, watch the class with headphones: there are low sounds and things that sound on the left and on the right. You don't need any instrument.

2. Listen to the song at the beginning without looking at the screen. How many different things are sounding at the same time? Write your number here: _____

3. **At the end of the class.** Go back to your number. If you counted four, you heard the layers; if you counted more, you also heard correctly: inside the rhythm there are three instruments.

While you watch the video

Fill in the table with each layer of the channel's song.

Layer	Which instrument plays it?	Low or high?	Short or long notes?	Its job
Rhythm				
Bass				
Chords				
Melody				

1. **In layers.** Who made a record playing six instruments by himself, one on top of another? Where and when? How does music software work today?

2. **Texture.** What is an accompanied melody? Why is the channel's song one?

3. **The piano roll.** What does it mean if a block is higher up? And if it is longer? Where does the name come from?

4. **The two notations.** In the score, where do you read a note's pitch? And its duration? What machine was the roll born for, and what is it called in music software today?

5. **If we take a layer away.** Which one did you miss most? What is lost without each one?

6. **A single wave.** Why do the four layers reach your ear as a single line?

7. **The cochlea.** Where do low notes vibrate? And high ones? Why does the bass also light up places towards the entrance? Does your ear hand you the four layers already separated?

8. **Bregman's six notes.** What do you hear when they play slowly? And when they play fast? What does your brain join together when the notes are so close together?

9. **The amadinda.** What do the two musicians play? What does the listener hear?

10. **The cues.** With what five cues does your brain group what it hears?

11. **The typical mistake.** What changed between accompaniment 1 and accompaniment 2? Did the melody change?

Do you hear it or build it?

Mark **H** if it is something that happens in the air or in the ear (physics and biology) and **B** if it is something your brain does when listening.

1. The four layers arrive added together, in a single wave.

2. Low notes make the far end of the cochlea vibrate.

3. The six fast notes are heard as two melodies.

4. The bass's harmonics are heard as a single note.

5. In the amadinda, two irregular rhythms are heard.

Listening challenge: which layer is missing?

Excerpt	The missing layer
1	
2	
3	

Reading

The first five notes of the melody, in treble clef: _ _ _ _ ___

Practice

1. In a song you like, write down the second at which each layer comes in.

2. If its intro doesn't have all the layers, say which ones are missing.

3. Explain in your own words how come you hear several things, if a single wave reaches your ear.

For exercise 1:

Layer	Second it comes in	How do you recognise it?
Rhythm		
Bass		
Chords		
Melody		

The solutions, at the start of the next Music class.

Keep thinking

Where is the song: in the wave that reaches your ear, in the score, or in your head? Give a reason for each one before saying what you think.

For the whole topic

Paper and a pencil and, if you can, some headphones. For the final workshop, a computer with a free and open-source program, MuseScore Studio (for Windows, macOS or Linux, in Spanish and with no account: https://musescore.org/en), which the video will teach you to use. If you don't have a computer at home, ask at your school: you can use theirs. What you make is yours: you don't need to show it or upload it anywhere.

All the music in the class was composed and synthesised by the channel; the hearing demonstrations follow Albert Bregman's experiments (1971 and 1990) and Greenwood's map of the cochlea (1990).

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